



KnowSelf

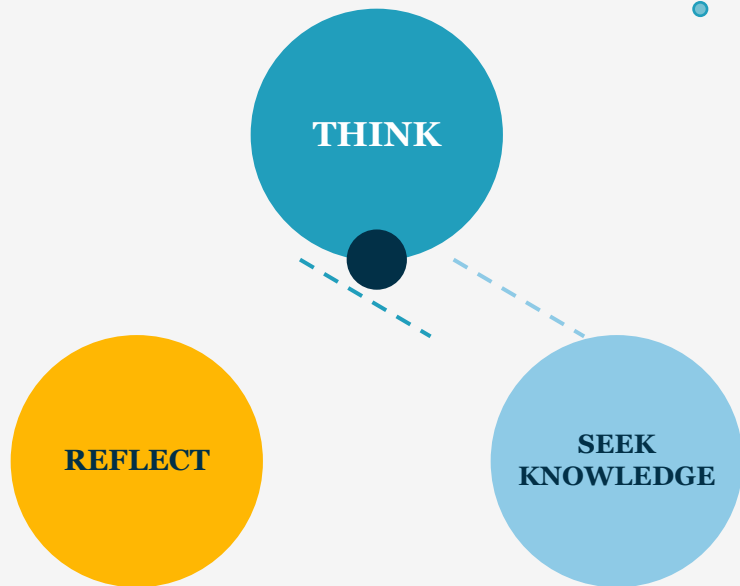
Agentic Knowledgeable Self-Awareness
for LLM-Based Agents

Teaching Agents When to Think,
Reflect, or Seek Knowledge

arxiv.org/abs/2504.03553

Enabling autonomous regulation of knowledge utilization
through situational self-awareness

Three Thinking Modes



Situational
Self-Awareness

Current Agent Training Uses “Flood Irrigation”

THREE EXISTING PARADIGMS — AND THEIR LIMITS

01

Trajectory Imitation

e.g., ReAct

Blindly copies gold trajectories regardless of agent capability at each step

02

Trial-and-Error Refinement

e.g., Reflexion

Injects external feedback indiscriminately — even when agent already knows the answer

03

Knowledge-Augmented Planning

e.g., KnowAgent, WKM

Floods ALL steps with domain knowledge — 100% knowledge usage even when unnecessary

THE CORE PROBLEM

Human Self-Awareness

Humans assess whether they need help before acting — calibrated, efficient, adaptive

VS

Agent Pattern-Fitting

Agents blindly apply learned patterns — leads to fragile planning, pattern collapse, high inference costs



Result: Fragile planning + unnecessary costs — No agent truly knows when it needs help

Three Situational Thinking Modes

At each decision step, KnowSelf evaluates:

<fast>

Fast Thinking

Criterion:

Agent predicts correct action directly

Description:

Proceed immediately — no overhead

Direct action execution

<reflect>

Slow Thinking

Criterion:

Initial action is wrong, but reflection fixes it

Description:

Self-correct through internal reasoning

Reflection then action

<knowledge>

Knowledgeable Thinking

Criterion:

Cannot correct even after reflection

Description:

Query external knowledge base

Knowledge retrieval then action

Special tokens trigger the appropriate inference mode at runtime

KnowSelf Pipeline: From Data to Self-Aware Inference

1 Self-Awareness Data Construction



- Agent self-explores environments
- Heuristic criterion classifies each step: fast / slow / knowledgeable thinking
- Special tokens inserted into trajectories

2 Two-Stage Training



- SFT — Supervised Fine-Tuning: learns initial self-awareness patterns
- RPO — Reward-based Policy Optimization: refines special token generation

3 Self-Aware Inference



- Agent generates special tokens during deployment
- Tokens trigger: direct action / reflection / knowledge retrieval from external KB

Data → Training → Inference

KnowSelf Correctly Reasons Where Frontier Models Fail

Task A: Heat & Place Mug

Task: "Put a hot mug in cabinet"

OpenAI-O1

WRONG
X

Tries to take an egg from the open microwave — ignores task goal

KnowSelf Llama-8B

CORRECT
✓

Uses <reflect> token → realizes: use the open microwave to HEAT the mug first

Mode used: Slow Thinking (Reflection)

Task B: Place Saltshaker

Task: "Put a saltshaker in drawer"

DeepSeek-R1

WRONG
X

Reflects that drawer is open → tries to close it first → action fails

KnowSelf Llama-8B

CORRECT
✓

Uses <knowledge> token → notes: place object without removing unrelated items

Mode used: Knowledgeable Thinking (Knowledge Retrieval)

KnowSelf (Llama-8B) outperforms both OpenAI-O1 and DeepSeek-R1 on these household planning tasks

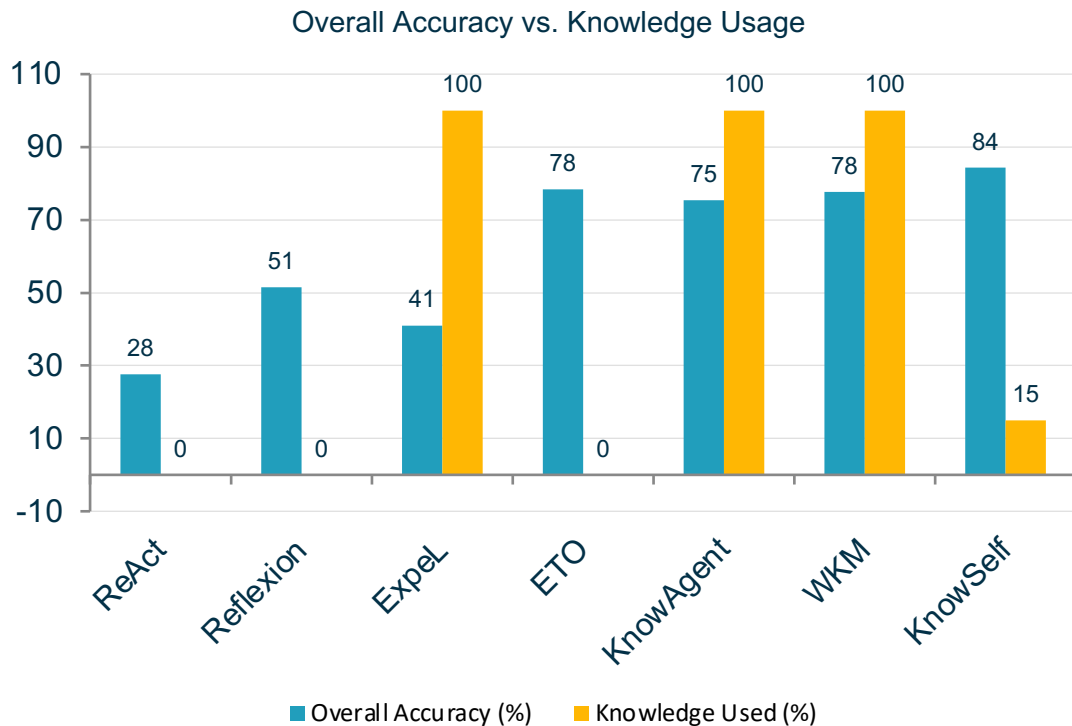
KnowSelf Achieves 84.33% on ALFWorld with Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf★	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf★	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Source: ALFWorld benchmark. Know% = proportion of steps using external knowledge.

15% knowledge → 84.33%
vs 100% knowledge → 77.61% (WKM)

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Edge



Llama-8B backbone results on ALFWorld benchmark.

84.33%

KnowSelf Llama-8B accuracy

Best overall on ALFWorld

15%

Knowledge queries used

vs 100% for KnowAgent/WKM

79.85%

KnowSelf Gemma-2B

Matches GPT-4o Expel at 2B scale

Key Takeaways: Self-Awareness Is the Missing Piece

01

Indiscriminate knowledge injection is wasteful — 100% knowledge usage can HURT performance vs selective use

02

Agents can learn situational self-awareness via lightweight heuristic labeling — no expensive human annotation needed

03

KnowSelf (Llama-8B) reaches 84.33% on ALFWorld — outperforming GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%)

04

Only 15–26% knowledge queries needed versus 100% for all competing knowledge-augmented methods

05

Paradigm generalizes across model scales: Gemma-2B reaches 79.85%, matching GPT-4o ExpeL

Future Directions

- Extend to multi-hop reasoning and longer-horizon tasks
- Apply three-situation framework to other agentic benchmarks

KnowSelf

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